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PHYSIOLOGICAL REVIEW

The whats and whens of sleep-dependent memory consolidation

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SUMMARY

Sleep benefits memory consolidation. The reviewed studies indicate that this consolidating effect is not revealed under all circumstances but is linked to specific psychological conditions. Specifically, we discuss to what extent memory consolidation during sleep depends on the type of learning materials, type of learning and retrieval test, different features of sleep and the subject population. Post-learning sleep enhances consolidation of declarative, procedural and emotional memories. The enhancement is greater for weakly than strongly encoded associations and more consistent for explicitly than implicitly encoded memories. Memories associated with expected reward gain preferentially access to sleep-dependent consolidation. For declarative memories, sleep benefits are more consistently revealed with recall than recognition procedures at retrieval testing. Slow wave sleep (SWS) particularly enhances declarative memories whereas rapid eye movement (REM) sleep preferentially supports procedural and emotional memory aspects. Declarative memory profits already from rather short sleep periods (1–2 h). Procedural memory profits seem more dose-dependent on the amount of sleep following the day after learning. Children's sleep with high amounts of SWS distinctly enhances declarative memories whereas elderly and psychiatric patients with disturbed sleep show impaired sleep-associated consolidation often of declarative memories. Based on the constellation of psychological conditions identified we hypothesize that access to sleep-dependent consolidation requires memories to be encoded under control of prefrontal-hippocampal circuitry, with the same circuitry controlling subsequent consolidation during sleep.

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Introduction

Amongst the different functions proclaimed for sleep, in recent research the importance of sleep for the consolidation of memories has received an upsurge of attention. Indeed, memory consolidation may be the only function that eventually can explain the loss of consciousness experienced during sleep, based on the fact that the brain uses basically the same limited neuronal network capacities for the acute conscious processing of information and its long-term storage. Acute processing and storing information might be mutually exclusive processes that cannot take place in the same networks at the same time.[1,2]

Memory function can be divided into three sub-processes: encoding, consolidation and retrieval. Encoding refers to the acquisition of the learning material which results in a newly built and initially labile memory representation. During consolidation, these fresh memory traces are strengthened and transformed into a more stable and persistent form and are integrated into pre-existing knowledge networks. Consolidated memories thus remain accessible at a delayed retrieval.

In 1924, Jenkins and Dallenbach were amongst the first to provide experimental evidence that sleep favors memory consolidation.[3] They systematically tested the retention of learned nonsense syllables over time and found that memory performance was better following a night of sleep than after an equivalent amount of time awake. Since then numerous studies examined the role of sleep for memory processing focusing on different memory tasks, different types of learning and retrieval, on the characteristics of post-learning sleep and of the subject sample. Overall, these studies impressively show that sleep promotes memory consolidation.[2,4–6] By now, it is basically beyond dispute that sleep can benefit memory consolidation. The central question is rather in which conditions sleep consolidates acquired information and in which it does not. Answering this question will eventually also help to enlighten the plastic processes that mediate the sleep-associated consolidation of memories. Whereas the mediating mechanisms are the focus of several previous reviews,[2,7–10] here we concentrate mainly on psychological conditions determining the efficacy of sleep for memory consolidation in humans (Figure 1). Specifically, we ask to what extent the effect of sleep on memory consolidation depends on i) the type of learning material, ii) the type of learning,

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Nomenclature

fMRI	functional magnetic resonance imaging
mPFC	medial prefrontal cortex
NREM	non-REM sleep
REM	rapid eye movement
REMD	REM sleep deprivation
SRTT	serial reaction time task
SWS	slow wave sleep

iii) the type of retrieval test, and iv) the type of sleep after learning. Finally v), we discuss findings in different subject populations.

Type of learning material

Declarative and procedural memory

Memory is commonly divided into a declarative and a non-declarative memory system.[11] Declarative memory is defined by memories accessible to conscious recollection, i.e., memories for events in a spatio-temporal context (episodic memory) and fact-based information (semantic memory). Non-declarative memory includes a heterogeneous collection of abilities resulting from experiences being not necessarily available for conscious recollection. Procedural memory for skills is the type of non-declarative memory most thoroughly studied with regard to the effects of sleep.

Verbal paired associate learning tasks have been most often employed to examine declarative memory consolidation during sleep. These tasks require the subject to learn a list of associated word-pairs and after sleeping or staying awake for several hours, cued recall is assessed. Usually, in these studies subjects who sleep after learning show better retention performance at cued recall testing than subjects who stay awake during the retention interval, thus providing consistent behavioral evidence that post-learning sleep enhances the consolidation of declarative memories.[12–14] Strong effects of sleep on memory consolidation have also been demonstrated for other declarative materials like nonsense syllables,[3,15,16] object locations,[17] short stories[18] and wordlists.[19,20] Declarative memory traces are highly susceptible to decay and forgetting, i.e., processes that counteract the presumed improved memory consolidation during sleep. Benefits of sleep for declarative memory consolidation, therefore, often express as a relatively diminished forgetting of the material at delayed retrieval testing.

Apart from increasing the amount of information (e.g., number of word-pairs) retained in memory, sleep also stabilizes these memories, as indicated by an increased resistance to interference.[21] Subjects learned a list of word-pair associates (A–B) before a 12-h retention interval of sleep and wakefulness, respectively. Prior to retesting, subjects learned an interfering list of word-pairs (A–C) which dramatically decreased the retention of A–B words in wake subjects while memory performance in sleep subjects was preserved, indicating that sleep makes memory traces resistant to interference.

The beneficial effect of sleep on declarative memory consolidation is assumed to rely on a process of system consolidation involving the reactivation of the initially labile memory traces in the hippocampal formation and their transfer from the hippocampus to neocortical sites for long-term storage.[1,2] A causative role of reactivations for the consolidation process was shown in studies indicating that experimentally induced reactivations of hippocampus-dependent visuo-spatial memories (by presenting associated odor cues during SWS after learning) indeed, substantially enhances these memories.[17] Moreover, functional magnetic resonance imaging (fMRI) revealed that sleep after learning, in comparison with post-learning wakefulness, leads to an enhanced

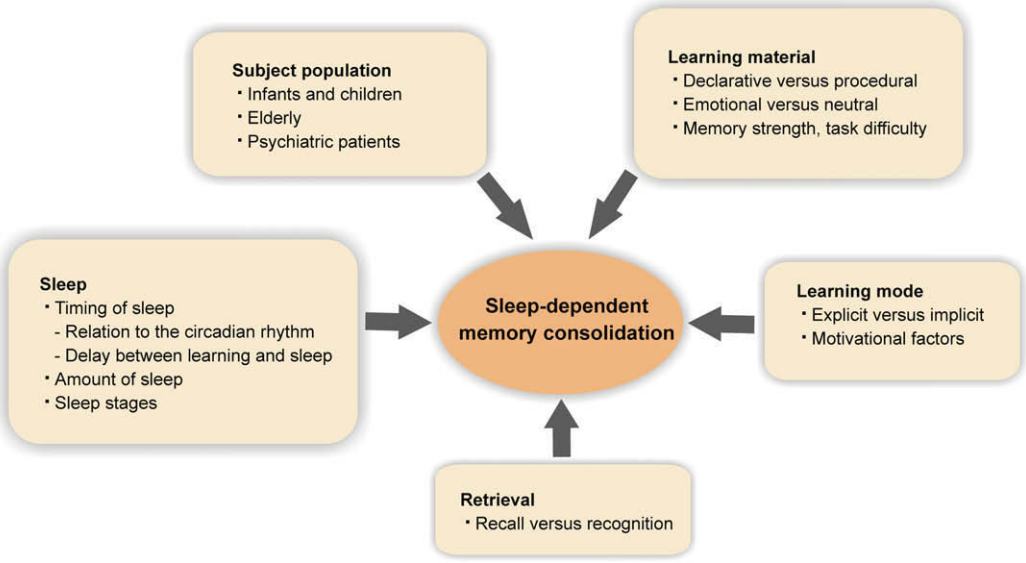


Fig 1. Critical conditions influencing the consolidation of memories during sleep, i.e., the type of learning material, the type of learning, the type of retrieval test, features of sleep and the subject population investigated. Distinct benefits of sleep were found for declarative and procedural as well as emotional materials. The enhancing effect of retention sleep is greater for weakly than strongly encoded associations, whereby the formation of weak associations can be a consequence of increased task difficulty. Sleep also strengthens the temporal sequence underlying an episode. Memories that are encoded explicitly and are linked to expected reward are particularly susceptible to sleep-dependent memory consolidation. In declarative memory, sleep benefits are more consistent with recall than recognition procedures at retrieval. Sleep supports memory consolidation independent of its circadian timing. Declarative memories seem to profit from rather short post-learning sleep periods (1–2 h). For procedural memories the benefit appears to be more dose-dependent on the amount of sleep occurring within ~1 day after learning. Slow wave sleep preferentially consolidates declarative memories whereas REM sleep preferentially benefits procedural and emotional aspects of a memory. Children's sleep hallmarked by high amounts of slow wave sleep supports declarative memory consolidation but does not produce an immediate gain of procedural skill. Elderly and psychiatric patients with sleep disturbances show impaired memory consolidation during sleep.

functional connectivity between the hippocampus and medial prefrontal cortex (mPFC) at a recall test 48 h after learning, and to enhanced activity in the mPFC and occipital cortex at a recall test 6 months later,[22] thus strengthening the neocortical representation of these declarative memories.

There is also strong evidence that sleep enhances consolidation of procedural memories.[23,24] For motor memory consolidation, the finger sequence tapping task and different adaptations of the serial reaction time task (SRTT) have been frequently studied. In finger sequence tapping tasks, subjects are required to repeatedly tap a given sequence of buttons on a key board. The underlying sequence is either explained before the training proper or displayed in front of the subject so that subjects are aware of the sequence while performing the task. In the SRTT (see Figure 2A,B), subjects are required to respond as fast as possible to the appearance of a target stimulus by pressing a spatially corresponding key. The appearance of target locations follows an underlying rule (grammar) that the subject typically remains unaware of.

Motor memory consolidation behaviorally expresses itself in two different ways: sleep can make motor skills, like declarative memories, more stable and resistant to interference. In addition, consolidation during sleep can produce a gain in skill, i.e., post-training sleep per se, in the absence of any overt training during the retention period, leads to enhanced speed and accuracy of skill performance at a delayed retesting. Walker et al.[25] proposed that stabilization occurs within 6 h after learning, independent of intervening sleep, whereas motor skill enhancement critically depends on sleep. However, recent studies challenged such strict dichotomy inasmuch as both measures were revealed to be sensitive to sleep-related consolidation. Enhancement in motor performance, although weaker, can be also present after retention periods of wakefulness.[23,26] Robust effects of sleep stabilizing finger motor sequence memories against interfering training on a different sequence have been shown by Korman and colleagues.[27]

Enhanced motor performance in the finger sequence tapping task after sleep has recently been shown to be related to characteristic changes in brain activation as revealed by fMRI.[28,29] Compared with a post-learning wake interval, sleep after training at a delayed retrieval test reduced activity in prefrontal, premotor and ipsilateral primary motor cortical areas, whereas activity was increased in parietal cortical and striatal areas.[28] Thus, sleep provides system-level consolidation also in the procedural memory system, by reorganizing neuronal motor representations toward enhanced efficacy. The reorganization likely arises from a reactivation of skill memories during sleep. Subjects who were trained on an SRTT showed enhanced activation and connectivity of training-related brain areas during post-training rapid eye movement (REM) sleep as revealed by positron emission tomography (PET).[30–32]

Not only motor skills but also perceptual skills like visual texture discrimination and sensory motor skills like mirror tracing have been shown to substantially improve across retention intervals filled with sleep, in comparison with both performance at training and with performance at retesting after corresponding wake retention intervals.[13,33,34] For visuomotor adaptation tasks, some but not all studies revealed performance gains as a result of sleep-dependent offline-consolidation.[26,35]

The declarative and procedural memory systems involve different brain structures. Declarative memory relies essentially on the hippocampus whereas procedural memories strongly rely on striatal and cerebellar function apart from neocortical contributions.[36,37] However, there is increasing evidence that these memory

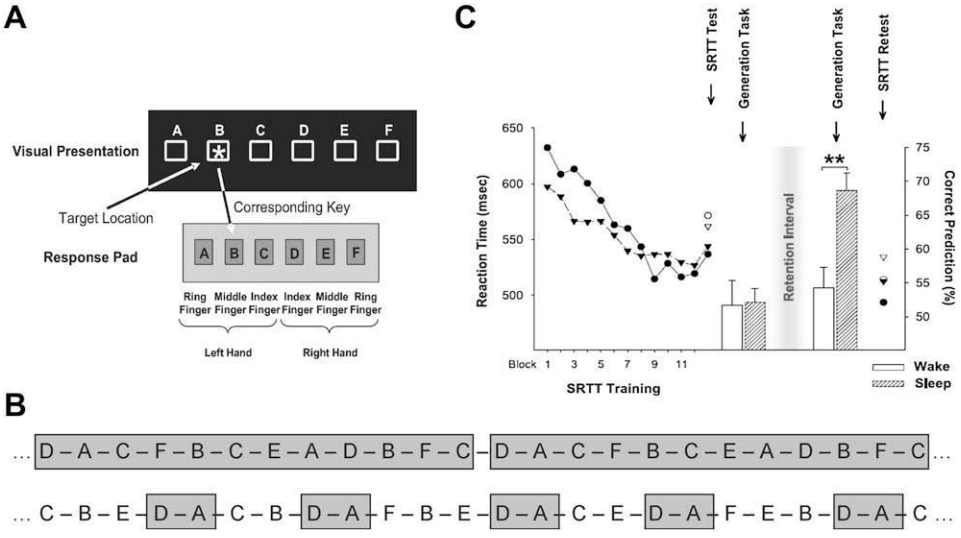


Fig 2. Effects of sleep on serial reaction time task (SRTT) performance. (A) The SRTT: subjects are presented with an array of horizontally arranged target locations (white boxes). They are instructed to react as fast and as accurately as possible to the occurrence of a target stimulus (white star) at one of these locations by pressing a spatially corresponding key on a response pad. (B) The sequence of target locations in the SRTT follows a specific set of rules (grammar) that can be deterministic (upper panel illustrates a repeating 12-element sequence) or probabilistic (lower panel, in this example, each two successive trials determined which of two possible target locations could legally follow. Thus, the successive positions "D" and "A" (gray fields) could be either followed by positions "C" or "F"). Implicit knowledge of the sequence grammar is indicated by the difference in reaction time to grammatical and random target positions that in test blocks are interspersed among the grammatical sequence positions. The SRTT can also be performed with the subject having explicit knowledge of the sequence grammar and, in this case, resembles the classical finger sequence tapping task. (C) Effects of sleep on explicit and implicit sequence knowledge in an SRTT (adapted from Fischer et al.[42]). Subjects were trained under implicit conditions on 12 blocks of a probabilistic SRTT followed by a test in which random target positions (empty symbols) were interspersed among the grammatical target positions (filled symbols). Performance is indicated in terms of reaction time (left *y*-axis) for a sleep (triangles) and a wake retention group (circles). To examine explicit sequence knowledge training was followed by a generation task in which subjects were asked to predict upon presentation of a certain target location the succeeding position of the target. Performance is expressed in terms of percentages of correctly predicted target locations (right *y*-axis). At retesting after the retention interval, subjects performed first on the generation task and then on the SRTT. Sleep induced explicit sequence knowledge: generation task performance did not differ from chance at learning. At retesting, however, subjects who had slept during the retention interval were distinctly superior in correctly predicting the respective next target location, compared with performance in the wake group which remained at chance level. Implicit knowledge of the sequence grammar (difference between respective empty and filled symbols at SRTT test) did not show the expected sleep-dependent improvement possibly due to the gain of explicit sequence knowledge interfering with skill performance.

systems are not as independent as originally assumed. Learning a task does not lead to isolated activation of only one of the memory systems. Thus, particularly in the initial stages of learning the hippocampus is also activated during procedural tasks.[38–40] In addition to co-activation, the memory systems interact and can even interfere. The preferential consolidation of declarative aspects of a task during sleep was found to concurrently suppress sleep-dependent gains in procedural skill on the same task.[41,42] Subjects who had gained explicit knowledge of the underlying sequence grammar in an SRTT at retesting after post-training sleep, did not show the expected sleep-dependent speeding of reaction times to grammatical cue positions (Figure 2C). However, in another study, introducing a declarative verbal paired associate task after training an SRTT did not prevent the development of offline gains in motor performance across subsequent sleep.[43] The interaction between the two memory systems during sleep-dependent consolidation is clearly in need of further study.

Emotional versus neutral material

In contrast with animal studies, studies of memory consolidation during sleep in humans have mainly employed neutral rather than emotionally arousing materials. In general, emotional events are remembered with greater accuracy and vividness than neutral ones (for review see Labar and Cabeza[44]). The brain area most important for the superiority of emotional memory is the amygdala which via its numerous projections modulates activity in other brain regions relevant to memory, particularly the hippocampus.[45,46] Amygdala activation during encoding enhances performance at retrieval,[47] and such amygdala activation may reoccur or persist after encoding, thereby influencing processes of consolidation.[48] Whether the post-acquisition brain state, i.e., sleep versus wakefulness differentially modulates the memory enhancing effect of emotional arousal has only been scarcely investigated. Amygdala activity is enhanced during sleep, especially REM sleep, in comparison to wakefulness,[49,50] possibly supporting synchronous activation between amygdala and hippocampus as well as neocortical sites underlying the long-term storage of emotional events.[51]

Based on a psychodynamic background, most early studies concentrated on the effects of REM sleep deprivation (REMD) on emotional memory processing. After REMD, subjects remembered less memory contents with emotional salience compared to subjects who slept normally whereas neutral memories were not affected.[52,53] However, overall results from REMD studies remain inconclusive,[54] because depriving subjects selectively from REM sleep by awakening them at first signs of REM sleep induces per se cognitive disturbances that confound later retrieval performance.[55]

The "early–late sleep" comparison represents an alternative experimental approach that excludes this confound (Figure 3A). Due to an underlying circadian rhythm, nocturnal sleep in humans is dominated by slow wave sleep (SWS) during the first half whereas REM sleep predominates during the second half. Comparing memory across retention intervals covering either the early or late period of nocturnal sleep, the involvement of SWS versus REM sleep can be investigated leaving retention sleep per se undisturbed. Adopting this paradigm, Wagner and colleagues[56] provided evidence for an enhancing effect of REM sleep on consolidation of emotional memories. Retention of emotionally arousing pictures was improved across an interval of late REM sleep-rich but not across early SWS-rich retention sleep, in comparison with the effects of corresponding wake intervals. A whole night of sleep improved recognition of emotional pictures compared to wakefulness whereas no memory-enhancing effect of sleep was found for neutral pictures in this study.[57] In an fMRI study, sleep-induced consolidation of emotional pictures at retrieval testing was associated with enhanced activity of hippocampal and medial prefrontal cortical (mPFC) regions, but not of the amygdala.[58] Also, functional connectivity between the hippocampus and mPFC was greater when acquisition was followed by

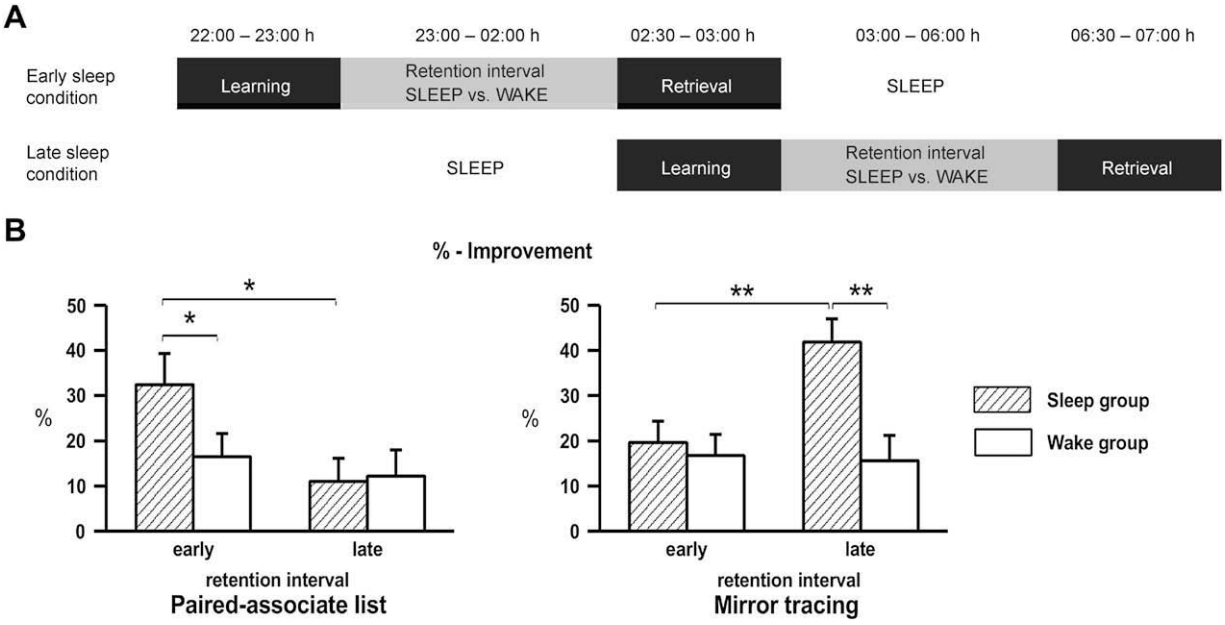


Fig 3. Contributions of early (SWS-rich) and late (REM sleep-rich) sleep to declarative and procedural memory consolidation. (A) Early-late sleep design. Due to an underlying circadian rhythm, SWS expresses most intensely during the first half of nocturnal sleep whereas REM sleep preferentially occurs during the second half. The involvement of SWS and REM sleep in memory consolidation can be investigated comparing retention intervals covering either the early or late period of nocturnal sleep leaving retention sleep per se undisturbed. (B) SWS-rich sleep during the early half of nocturnal sleep specifically enhances declarative memories (here: verbal paired associates) compared to the late half of nocturnal sleep and wake retention intervals covering corresponding nocturnal periods. Procedural memory (here: mirror tracing skill) was selectively enhanced across retention intervals covering the late (REM sleep-rich) half of nocturnal sleep while retention performance remained unaffected across early sleep or wakefulness. Retention performance is indicated as percentage of improvement relative to initial learning performance. Adapted from Plihal and Born.[13]

sleep compared with wakefulness. These results suggest that amygdalar activation during retention sleep enhances the hippocampo-neocortical dialogue during sleep thereby benefiting the consolidation of episodic aspects of the memory. Whether the emotionality of the memorized experience per se is changed by retention sleep is not clear.

Together, there is consistent evidence that sleep and especially REM sleep supports emotional memory consolidation. Patterns of emotional arousal that are induced during learning via amygdalar circuitry possibly become reactivated during REM sleep[59] thereby strengthening memory traces and connectivity within hippocampo-neocortical networks.

Memory strength, depth of encoding and task difficulty

Memory representations can differ greatly in the strength of the underlying associations. Although discussed already in earlier reviews,[18,60] the dependence of sleep-associated memory consolidation on the strength of acquired associations has rarely been systematically tested. Available data suggest that benefits from sleep are greater for weaker than stronger traces.

The strength of a memory can be principally manipulated by varying the depth of encoding, for example by increasing the number of learning trials. Also, with greater strength of underlying associations the subject typically experiences a task as less difficult, i.e., performance on difficult tasks relies on relatively weak associations. Examining declarative memories for word-pairs Drosopoulos et al.[61] showed that post-learning sleep produced a distinctly greater memory benefit for word-pair lists learned to a criterion of 60% correct responses (at an immediate recall test during the learning phase) than for lists learned to a criterion of 90% correct answers. The same study varied encoding depth also by introducing interference. In an interference paradigm, the strength of word-pair associations (A–B) learned first becomes substantially weakened if a second, interfering list of word-pairs (A–C) is learned shortly afterwards (due to retroactive interference). Sleep after learning, compared with wakefulness, distinctly enhanced recall of weakly associated words from the first-learned list (A–B) whereas the improving effect of sleep on the second-learned list (A–C) remained non-significant.[61,62] This pattern cannot be explained solely by ceiling effects because memory retrieval after retention sleep was far from perfect also for the strong associations.

In a related study, Ellenbogen and colleagues[21] adopted the same A–B, A–C interference learning paradigm as used by Drosopoulos et al.,[61] however aiming at a different issue, i.e., to examine whether sleep-associated consolidation makes memories more resistant to retroactive interference. Subjects learned the first list (A–B) to a criterion of 100% correct (thus forming strong associations) and learning of the second list (A–C) took place not before, but after 12-h retention intervals filled with sleep or wakefulness, and immediately before testing recall of the first-learned list (A–B). Interference learning strongly impaired recall of the first (A–B) list in wake subjects whereas A–B list recall in subjects who had slept after learning remained unaffected, suggesting that strong memory associations indeed benefit from sleep in that they are transformed to a more stable form resistant to interference.

Focusing on procedural memory, Kuriyama et al.[63] varied the difficulty of a finger sequence tapping task by varying the number of sequence elements and between uni- and bimanual performance. Post-learning sleep induced the greatest performance gain for the most difficult task. Analyses of single transitions between sequence elements likewise revealed the greatest overnight improvement in speed for the transitions that were slowest (i.e., most difficult) at training.

Together these results speak for the notion that sleep enhances weak associations in memory to a greater extent than strong associations, though strong associations might also benefit from sleep by becoming more resistant to retroactive interference. However, two recent studies report divergent results, i.e., greater sleep benefits for strong memories. Hauptmann et al. using a repetition priming task found delayed gains after 24 h only in subjects whose performance leveled off and had reached an asymptotic plateau during training.[64] Unfortunately this study lacked a wake control group. In a study by Tucker and Fishbein only subjects who performed well at learning of declarative tasks (word-pairs, maze learning, complex figures) showed a benefit in retention after a nap, in comparison with a wake control condition, whereas no difference was observed for low-performers.[65] Assuming that high-performing subjects had encoded stronger associations, these findings are in contrast to the hypothesis that sleep preferentially strengthens weak associations. However, these outcomes might also reflect an individual trait, i.e., sleep could be generally less effective in low-performing individuals.[66,67]

Temporal order

An episode is characterized by a specific temporal order of events, with the storage of temporal sequence information relying essentially on hippocampal function.[68] To date, only one study has examined whether sleep enforces the consolidation of the temporal sequence underlying a specific episodic memory, beyond strengthening the memory for the single events.[69] In this study, subjects learned triplets of words (A–B–C) before retention periods of sleep and wakefulness. At retrieval a cued recall test was performed that required the subject to retrieve the association either in forward manner (e.g., respond to cue word A with word B) or backwards (e.g., respond to cue word C with word B). Post-learning sleep selectively enhanced forward associations whereas backward associations (although generally weaker already at learning) remained completely unchanged. These findings speak strongly for a strengthening effect of sleep also on the temporal sequence information contained in an episode, and are well in line with animal studies observing a neuronal "replay" of episodes (experienced in the wake phase) mainly during subsequent SWS[70–73] but also during REM sleep.[74] This replay activity which is considered a neuronal phenomenon underlying the consolidation process in hippocampal and neocortical networks, follows the same temporal order as during prior learning.[70–73,75] Neuronal reactivations can also occur in a backward direction, but these have been primarily observed during waking.[76,77] The sleep-associated enhancement of temporal sequence information may thus rely specifically on neuronal reactivations temporally ordered in a forward direction, which occur preferentially during sleep.

Type of learning

Explicit versus implicit learning

Learning can be explicit, i.e., the subject is aware that something is learned, or implicit, with no awareness that something is learned. Whereas the acquisition of declarative memories is generally explicit, the acquisition of procedural skills typically involves both implicit and explicit learning. Learning how to dance, for example, relies on explicit instruction of the sequence of steps apart from repetitive practice. Under experimental conditions, explicit learning is often intentional, i.e., the subject adhering to explicit task instructions intends to memorize certain information, but learning can be also incidental without the subject's awareness that something is to be learned. To what extent intentionality of

learning can enhance sleep-associated consolidation of memories has not been examined so far.

The SRTT can be used to study effects of explicitness of learning on sleep-associated consolidation (Figure 2). Subjects typically perform on the SRTT like on a reaction–time task, i.e., they are required to rapidly respond to specific cues by pressing corresponding buttons, without knowledge that there is an underlying sequence grammar, i.e., implicitly. However, the SRTT can also be performed with explicit knowledge by informing subjects that there is an underlying sequence grammar. With deterministic sequences, the subjects may, to a certain extent, even be aware of the exact transitions of cue positions in the sequence. In this case, the SRTT basically resembles the classical finger sequence tapping task that requires the subject to repeatedly tap an explicitly given sequence. Using explicit finger sequence tapping tasks studies consistently revealed distinctly greater performance improvements across retention periods filled with sleep compared to wakefulness.[23,24,27] In contrast, using implicit forms of the SRTT, the gains in speed (to grammatical cue positions) after post-learning sleep were not always revealed to be superior to that after the wake control condition, in particular when probabilistic sequence grammars were used to strictly exclude development of any explicit sequence knowledge during training.[78,79] Comparing directly overnight gains in explicit and implicit SRTT performance, two studies revealed significantly larger gains in speed after sleep than wakefulness only when the subjects were aware of the underlying (deterministic) sequence,[80,81] whereas gains under implicit conditions were comparable for the sleep and wake control conditions. However, sleep induced significant overnight gains under implicit conditions when subjects were required to respond to contextual cues, i.e., specific colored stimuli, whose occurrence was correlated with the underlying sequence unknown to the subject.[81] The processing of contextual information is known to rely on hippocampal function,[82] possibly accounting for the sleep benefits in this context-associated version of the implicit SRTT.

In addition to preferentially consolidating hippocampus-dependent and explicitly encoded memories, sleep appears to promote the transformation of implicitly acquired memory traces into explicit knowledge. Compared with a wake retention control condition, subjects after a period of retention sleep were more able to explicitly generate the sequence underlying an SRTT which they had implicitly learned before sleep[42] (Figure 2C). In fact, subjects after the wake retention interval did not develop any explicit sequence knowledge and remained at chance level when trying to generate the sequence. Others showed that sleep, and especially slow wave sleep, after practicing a more complex cognitive problem solving task (i.e., a number reduction task) promotes the gain of (explicit) insight into the hidden structure embedded in the task that was not seen before sleep or after corresponding retention periods of wakefulness.[41,83]

Overall these studies support the view that sleep preferentially consolidates explicitly learned materials. Being aware of the specific task stimuli to be learned, i.e., explicit learning is known to involve activation of the hippocampus.[84,85] Thus, engagement of the hippocampus at learning might be a critical factor in making a memory trace susceptible to sleep-dependent consolidation processes independent of whether this is due to the kind of task (declarative, with context associations, etc.) or type of learning (explicit). Likewise, the hippocampal engagement at learning may be critical to sleep transforming implicitly encoded aspects of a task into explicit knowledge.

Motivational factors

In everyday life great amounts of information are encoded that eventually are not of any relevance for the individual. Here the intriguing question arises whether sleep-associated consolidation preferentially benefits memories that are behaviorally relevant and in any way associated with future reward. The issue was examined in a recent study, associating monetary reward with one of two previously trained finger tapping sequences (Fischer et al., submitted). Subjects learned successively the two sequences before 12-h retention intervals of nocturnal sleep and daytime wakefulness, respectively. Immediately after training the sequences it was announced to the subjects that they could earn extra money if at later retrieval testing they showed optimal performance on either the first- or second-trained sequence, depending on the experimental condition. Post-training sleep compared to diurnal wakefulness enhanced overall finger sequence tapping performance at retest. However, the sleep-dependent gain was significantly greater on the sequence that had been associated with monetary reward, regardless of whether this sequence was trained first or second. Importantly, to control for immediate effects of motivation on retrieval, shortly before retest, subjects were informed that the monetary reward would *not* depend on only the one previously announced sequence, but on overall performance on both sequences. Reward anticipation did not influence memory performance after a retention interval filled with wakefulness. The findings represent first evidence that motivational factors contribute to whether or not a memory trace gains access to sleep-dependent consolidation.

Type of retrieval test

Especially for declarative memories, the type of retrieval test might determine whether a benefit from post-learning sleep compared to wakefulness is revealed. Experimentally, retrieval of declarative materials is tested using either "recall" or "recognition" procedures. Recall is the ability to remember a previously encountered stimulus in the absence of that stimulus, i.e., the to be remembered stimulus has to be *generated* by the subject himself, either without any specific cues given (free recall) or in response to a cue previously paired with that stimulus (cued recall). Recognition is the ability to decide in the presence of a stimulus whether this stimulus was previously presented or not, i.e., the learned stimulus has to be *recognized* by the subject. Recognition can express as "recollection" which refers to a remembering with a sense of "re-living" the stimulus including detailed spatio-temporal context information of its presentation, and as feeling of "familiarity" referring to simply knowing that the stimulus was previously encountered without the retrieval of specific context information.

Most studies that report beneficial effects of sleep on declarative memory consolidation used cued recall procedures.[12,86–89] Cued recall performance was consistently enhanced after post-learning periods of sleep, especially of SWS in the first half of the night, compared to respective wake intervals.[13,21,22,61,90,91] Free recall procedures likewise revealed a pronounced superiority of retention intervals filled with sleep as compared to wakefulness, although free recall was assessed in only a few studies.[15,16,20]

Recognition memory has also been scarcely examined, and these studies report only small effects[92,93] or even no beneficial effect of sleep on overall recognition performance.[57,94,95] Two studies found sleep effects only for recollection after SWS-rich sleep, but not for familiarity judgments.[95,96] Only one study revealed effects of retention sleep for familiarity judgments, and here only for emotionally arousing but not neutral materials.[57]

In combination these studies indicate that cued and free recall are better suited to unravel the effects of sleep on declarative memory consolidation than recognition tests. For correct recall, as compared to recognition, the subject himself reinstates the item to

be remembered. This process is thought to reflect basically the accessibility of a memory.[97] With enhanced recall, the target item is embedded in a richer network of neighboring associations providing possible access, and this could be the consequence of sleep promoting the integration of newly acquired memories into the network of pre-existing long-term memories. In fact, it has been previously proposed that sleep strengthens memories mainly in a system consolidation process that integrates and interlinks newly encoded memories with pre-existing knowledge networks.[98]

In recognition, on the other hand, the target stimulus need not be generated by the subject but is already sufficiently activated through its presentation. Whether it is recognized or not mainly depends on the strength of that particular memory to exceed a certain threshold. Additionally, recall and recognition differ in their underlying neuroanatomical structures. Recall is known to involve hippocampal function whereas hippocampal contributions to recognition, and especially to familiarity judgments, appear to be negligible.[99–102] On this background, recall revealing greater and more consistent sleep-dependent improvements of memory than recognition procedures, further corroborates the view of a preferential consolidation of hippocampus-dependent memories during sleep.

Type of sleep

Timing of sleep

Timing of sleep in the circadian rhythm

Sleep is in part a circadian phenomenon that in humans expresses most intensely during night time. Hence, any interaction of sleep and associated consolidation processes with the circadian rhythm cannot be excluded. Surprisingly few studies have examined the effect of the timing of sleep during the 24-h cycle on memory consolidation. Finger sequence tapping improved to the same extent after 8 h of sleep during the night and 8 h of daytime sleep and was in both cases significantly better than after corresponding periods of wakefulness.[23] Also, the improvement in retention of nonsense syllables and short stories was comparable after naps of 2 h placed either in the morning or in the afternoon, although the morning naps contained significantly more REM sleep whereas the afternoon naps contained more SWS.[103] In a study by Koulack,[93] 4 h of sleep placed either in the early morning or in the evening proved equally effective in enhancing recognition of words learned before retention periods of sleep and wakefulness. Together with several midday nap studies reporting beneficial effects on memory,[14,104,105] these studies strongly speak for the notion that it is sleep per se that promotes memory consolidation, independent of the time of day it occurs.

Time between learning and sleep

In educational settings, i.e., at school or university, most of new learning takes place during the day and only rarely in the evening hours before bed time. In contrast, in experimental investigations of sleep-associated memory consolidation subjects most often learn in the evening and sleep shortly thereafter. Astonishingly, it is still unresolved whether sleep must occur within a specific time-window after learning to enhance memory consolidation.

For declarative memory, the beneficial effect of sleep on the consolidation of word-pairs appeared to be greater when sleep followed learning within less than 3 h compared to longer delays of more than 10 h.[91] An earlier study also using word-pair learning[88] likewise found immediate sleep more effective than sleep delayed by 12 h. Unfortunately, these and similar studies remain basically inconclusive, because they do not take into account that with delaying sleep, subjects stay awake for a longer time during which the newly encoded memories are subjected to processes of forgetting and interference. Thus, the enhancing effect of delayed sleep on a memory cannot be estimated unless the amount of forgetting across the preceding wake time is properly measured.

For procedural finger sequence tapping the improvement in skill after immediate sleep[106] does not quantitatively differ from that observed after a sleep period of equal length which is delayed by 10–12 h.[24,25,27,107] Sleep periods immediately following learning and sleep periods delayed by 12 h post-learning revealed sleep-dependent improvements of comparable size in the visual texture discrimination task and in perceptual learning of an artificial language.[34,108] However, only marginal improvements in skill were found when retention sleep was delayed by more than 24 h[23,109] indicating that the emergence of gains in procedural skill requires sleep to occur within the same day of training.

Amount of sleep

Memory consolidation could be affected by the amount of sleep in two possible ways: (i) a certain minimum time in sleep is required for enhanced consolidation to be expressed in an all-or-none way with longer sleep duration producing no additional enhancement. (ii) Sleep benefits memory consolidation proportionally in a dose-dependent manner; the more sleep the greater the benefit. So far, this issue has not been studied systematically. However, a preliminary answer to this question might be derived from comparing studies that examined basically three types of interval lengths: naps with short durations of 30–120 min, sleep during night halves of 170–240 min duration and whole night sleep of 7–8 h.

Consolidation of declarative memories (mostly for word-pairs) has been consistently revealed to be enhanced after a whole night of sleep[22,61,88,91] as well as after the first (SWS-rich) half of nocturnal sleep,[12,13,86,87,90,110] in comparison with effects of a corresponding wake retention interval of equal length. Three out of four studies investigating the effects of a short nap on word-pair retention reported significantly better memory performance after sleep compared to wakefulness[14,65,111] whereas in one study this difference failed to reach significance.[112] Even an ultra short nap of about 6 min improved word retention compared to a wake control group, though a longer nap of 35 min was superior to the ultra short nap.[20] Given that the magnitude of the sleep-induced enhancement in memory for word-pairs in these studies seemed roughly comparable for whole nights and early night halves of sleep and even for the naps, there might indeed be a certain amount of only 1–2 h of sleep required to achieve optimal consolidation of declarative memories. Within these first 1–2 h, sleep benefits might be dose-dependent on the amount of sleep, with subsequent additional sleep providing no further benefit.

As to procedural memory, visual texture discrimination and finger sequence tapping was studied using different durations of retention sleep. Visual discrimination skill substantially improves after a whole night of sleep[33,34,109] and also after a 3-h interval of early nocturnal sleep.[33] However, the improvement across the entire nocturnal sleep period was about three times greater than that after early sleep alone,[33] and the total sleep time was significantly correlated with the overnight improvement in performance.[34] Short naps of 60 or 90 min also improved visual texture discrimination performance, but only if the nap consisted of both SWS and REM sleep.[105] The magnitude of the improvement was comparable to that seen over a full night of sleep by Stickgold et al.[34] but lower than that revealed overnight by Gais et al.[33] Finger sequence tapping performance improves after a whole night of post-training sleep,[23,24,27,106,107] as well as after an intervening nap

of 60–90 min.[27,104] However, the gains induced in the nap studies overall appeared to be less robust. Retesting of visual texture discrimination and finger sequence tapping after more than one night of post-learning retention sleep revealed small, but in some cases significant, additional improvements of skill after 2–7 nights of sleep.[24,25,107,109] Another study found no further improvement after two nights compared to one night of sleep.[23] Overall, the data suggest that the amount of sleep benefits procedural memory consolidation, within a certain range, in a dose-dependent manner, with more sleep resulting in a greater benefit.

However, studies with varying durations of the sleep period are difficult to compare because the amount of post-learning sleep in these studies is always confounded either with i) the length of the retention interval, ii) the amount of wakefulness between learning and retrieval, and iii) the proportion of different sleep stages. Typically the retention interval in studies examining whole nights of sleep is longer than in nap studies, and the longer retention interval is associated with increased rates of forgetting, i.e., processes possibly counteracting the effect of sleep. Introducing retention intervals of equal length with different amounts of sleep, on the other hand, excludes confounding effects of forgetting but implicates different amounts of wakefulness and associated interference between learning and retrieval. Finally whole night sleep is characterized by a unique distribution of sleep stages differing strikingly from the structure of sleep during just one night half or midday naps.

Sleep stages

Sleep is characterized by the cyclic occurrence of REM sleep and non-REM sleep comprising SWS (sleep stages 3 and 4) and lighter sleep stages 1 and 2. Two main hypotheses have been proposed regarding the role of sleep stages in memory consolidation. The *dual process theory* assumes that the specific sleep stages support consolidation of different types of memories. SWS supports declarative memory consolidation whereas REM sleep does so for procedural memories.[13,113–115] The *sequential hypothesis*, on the other hand, proposes that sleep benefits memory optimally through the cyclic succession of both SWS and REM sleep. The original version of this hypothesis assumed that SWS functions to weaken non-adaptive memory traces whereas REM sleep re-stores the remaining traces.[116,117]

The dual process hypothesis received support mainly based on the early-late sleep comparison, i.e., an approach comparing effects of retention intervals covering the first (SWS-rich) or the second (REM sleep-rich) half of nocturnal sleep (Figure 3A). SWS-rich early sleep was consistently found to support consolidation of hippocampus-dependent declarative memories, i.e., for word-pairs[12,13,86,87] and spatial relations,[114] as well as for memories explicitly recollected in recognition tasks,[95,96] whereas REM sleep benefited non-declarative types of memory like priming,[114,118] and memories for visuo-motor (mirror tracing,[13] Figure 3B) as well as cognitive skills (logic task solving[119]). However, this dichotomy does not fit all results. Several non-declarative tasks, like visual texture discrimination[33] and rotation adaptation,[35] are also supported by SWS whereas REM sleep in some instances seems to benefit aspects of declarative memory,[120] especially if emotional materials are used.[56,94] Earlier studies using REM sleep deprivation likewise pointed to an involvement of REM sleep in the consolidation of short stories and sentences.[18,19] However, REM sleep deprivation procedures have been criticized because of confounding effects of stress caused by repeatedly arousing the subject.[55] The early–late sleep design excludes this confound but still has some limitations. Thus, early and late sleep conditions differ with respect to circadian phase and sleep homeostatic pressure. Further, this design does not allow for an adequate assessment of contributions of stage 2 sleep to memory consolidation.

Support for the sequential hypothesis derives mainly from studies introducing disruptions of the natural cyclic sequence of SWS and REM sleep by awakenings from REM sleep.[117,121] These studies show that cycle disruptions impair memory retention but, like REM sleep deprivation, disrupting the sleep cycle can be criticized based on the stress-related confounds induced by this procedure. Nevertheless, several studies of undisturbed sleep, using correlation analyses, have suggested that the overnight gain in performance on a procedural visual discrimination task is in fact greatest when SWS *plus* REM sleep occur in succession during post-learning sleep.[33,34,105]

However, the classification of sleep into SWS and REM sleep is definitely too crude for adequately describing the mechanisms of sleep-dependent memory consolidation. Recent studies have concentrated on more fine-grained analyses of distinct polysomnographic phenomena, like sleep spindles and slow-wave activity (the latter including the <1 Hz slow oscillation). Increases in spindle activity, numbers and density of spindles and in the duration of stage 2 sleep are prominent after learning of verbal memory tasks[90,122–124] as well as after visuo-spatial learning[125] and learning of complex procedural motor skills.[104,126,127] Increases in slow-wave activity have been observed after learning a procedural rotation adaptation task and declarative word-pair learning.[35,128] Increases in slow wave activity after rotation adaptation learning were restricted to the motor cortical areas required for task acquisition. Inducing sleep slow oscillations by transcranial application of slowly oscillating potentials intensifies SWS and enhances consolidation of declarative memory.[129] Suppressing SWS with non-arousing tones, on the other hand, prevented the expression of overnight performance gains in visual texture discrimination skill.[130]

In sum, there is considerable evidence that SWS preferentially supports declarative memory consolidation whereas REM sleep seems to benefit specifically non-declarative and emotional aspects of a memory. However, the classical classification of sleep stages is too crude for an adequate reflection of the complexity of sleep-associated processes contributing to memory consolidation, which requires more fine-grained analyses of underlying electrophysiological and neurochemical processes. Also, for procedural memory consolidation, involvement of specific sleep stages has been proposed to depend on the subjects' initial skill level.[127] According to this view, tasks requiring the formation of novel skills involve REM sleep whereas the "fine-tuning" of existing skills requires stage 2 sleep.[131]

Type of subject population

Most research on sleep-dependent memory consolidation has been done in young healthy students. Examinations of subjects differing in various physiological and psychological aspects from this population (e.g., in age or health) are scarce although this kind of research can essentially contribute to our understanding of the function of sleep in memory processing.

Infants and children

The early developmental period is characterized by a great extent of behavioral and brain plasticity determining the child's capacity to easily acquire huge amounts of new facts and skills. In parallel, sleep in early infancy is characterized by huge amounts of SWS, declining with age and reaching already fairly low levels in midlife.[132] Indeed, there is first evidence that, in parallel with these alterations, the effects of sleep on memory consolidation differ in children and adults. Gomez et al.[133] in a learning phase familiarized 15-month old infants with auditory strings of words of an artificial

language. The infant's orienting response, i.e., turning his/her head towards familiar and unfamiliar strings, was used to asses delayed retrieval. Compared to a non-napping control group, children who had napped after learning appeared to be more able to abstract a rule-like pattern underlying the strings of words. However, signs of correct remembering of the presented words were enhanced in the wake group.[133] Two recent studies in children aged 6–8 and 9–12 years, respectively, showed that effects of sleep on the consolidation of hippocampus-dependent declarative memories are comparable to those in adults. Both children and adults distinctly benefited from periods of nocturnal sleep after acquisition of word-pairs and spatial memories (2D-object locations).[134,135] However, in contrast to adults, children did not show the expected sleep-dependent gain in speed or accuracy in procedural motor tasks (SRTT and finger sequence tapping task).[79,135] Improvements in skill appeared to be even greater across the wake retention interval. Yet, an immediate lack of overnight gains does not necessarily exclude an improving influence of post-training sleep on the long term: in young zebra-finches learning a song, song performance deteriorated across nocturnal sleep. However, the birds that showed strongest post-sleep deterioration achieved a better final song imitation at the end of the 3-month study epoch.[136]

To summarize, sleep in children like in adults strengthens declarative memories. Whether this effect is even stronger in children due to the preponderance of SWS remains to be explored. A direct comparison of sleep-dependent gains in children and adults is precluded due to differences in the tasks used to examine memory in both age groups in the respective studies. On the other hand, unlike adults, children do not display sleep-dependent overnight gains in two studies of procedural skill. This is surprising given that the neuroanatomical structures underlying procedural memory formation mature earlier than those contributing to declarative memory formation.[137,138] The results indicate that procedural memories are differentially processed in children.

Elderly

Sleep in the elderly is hallmarked by a distinct reduction in SWS and slow oscillations. However, sleep is also more fragmented, sleep latency is increased, total sleep time and time in REM sleep as well as REM density and sleep spindles are decreased.[139] In parallel, memory function deteriorates suggesting that both disturbances of sleep and memory might be interdependent. Middle aged subjects (aged 48–55 years) remembered less word-pairs after a period of early nocturnal sleep than young subjects (aged 18–25 years).[134] The older subjects spent less time in SWS and the amount of SWS during the early sleep period highly correlated with retention performance, i.e., the less SWS the worse memory recall.

Procedural memory consolidation during sleep was also found to be disturbed in older subjects. Whereas performance in an SRTT was comparable between old and young subjects during learning, unlike younger subjects the older subjects did not show any substantial increase in motor speed to the grammatical cue positions across the nocturnal sleep period,[140] an effect possibly related to a general decrease in sleep spindles in older subjects.[141] Spindle density following acquisition of a motor task increased significantly in younger but not in older subjects.[142] However, REM sleep deficits could also contribute to declining procedural memory consolidation in the elderly. Increased phasic REM sleep after administration of a cholinesterase inhibitor (increasing cholinergic tone) significantly enhanced sleep-dependent gains in procedural mirror tracing skill in the elderly.[143]

Apart from decreases in SWS, spindles and REM sleep, structural changes in hippocampus and neocortex could contribute to the age-dependent decline in sleep associated memory consolidation.[144] However, the reactivation of hippocampal ensemble activity patterns during SWS after spatial learning, which is considered to promote the consolidation of declarative memories, was not found to be changed in old rats.[145,146] The elderly also show enhanced cortisol concentrations during early sleep[147] which can impair consolidation of hippocampus-dependent memories during sleep.[148] The cholinergic tone, on the other hand, is distinctly decreased in the elderly,[149] possibly accounting for reduced sleep-dependent gains in procedural memory consolidation.[143] Overall, available data indicate diminished capabilities of consolidating memory during sleep in the elderly which are related to distinct changes in sleep architecture and associated transmitter and endocrine regulation.

Psychiatric patients

Investigations of sleep-dependent memory consolidation in psychiatric populations are scarce, although sleep and memory deficits occur jointly in a number of psychiatric diseases. Patients with chronic insomnia showed impaired consolidation of declarative memories during sleep in conjunction with diminished amounts of SWS.[150] After retention sleep, insomnia patients remembered significantly less word-pairs compared to learning performance before sleep, whereas memory performance even increased after sleep in healthy controls. Procedural skill memory (mirror tracing) was not differentially affected by sleep in patients and controls in this study. Yet, in another pilot study overnight gains in mirror tracing performance were revealed to be distinctly lower in insomnia patients than in matched controls.[161]

Depressive patients are conspicuous because of their enhanced memory for emotionally negative material[162] in conjunction with signs of disinhibited REM sleep, i.e., shortening of REM sleep latency and increased REM density. There is one study on depressed patients[151] indicating that patients who show better overnight retention of complex figures (of the Rey Osterrieth Complex Figure Test) sleep longer and show more REM sleep than patients with low retention performance. Procedural memory consolidation (for mirror tracing skill) was not correlated with sleep parameters. Unfortunately, emotional memory consolidation was not tested and the study also lacked a healthy control group.

In borderline personality disorder no signs of altered memory consolidation in declarative (word-pairs) as well as procedural memory (mirror tracing) during sleep were revealed although these patients showed increased REM sleep duration compared with healthy controls.[152] Schizophrenia is associated with reduced SWS and reduced REM sleep latency.[153,154] Schizophrenic patients were found to show decreased retention of declarative memories (complex figures, spatial locations) after sleep compared to healthy controls.[155] Interestingly, the memory impairment was correlated with reduced amounts of SWS and reduced sleep efficiency. Sleep-dependent gains in procedural skill (mirror tracing) were comparable between patients and healthy controls in this study,[155] whereas in another study[156] schizophrenic patients did not show the typical overnight gain in procedural finger sequence tapping skill.

Overall, these first data in patients support the notion of an interplay between sleep and memory processing. However, probably due to the great heterogeneity of the patient groups and the great number of confounds (medication, comorbidities, drug intake) typical for these studies, data so far do not give a clear picture as to specific links between alterations of sleep architecture and aspects of memory consolidation in these patients.

Conclusion

This survey aimed to specify psychological conditions that influence the consolidation of memories during sleep (summarized

in Figure 1). Although sleep benefits memory consolidation in a wide range of conditions, this review shows that the consolidating effect is not revealed under all circumstances but is linked to a specific constellation of conditions which eventually may help to clarify the nature of this phenomenon. Although speculative, in combination these factors seem to support the idea that sleep preferentially strengthens memories that were encoded under explicit conditions with essential contributions of a prefrontal cortical-hippocampal network that integrates declarative, procedural, emotional and motivational aspects within a conscious process of encoding. Activation of the hippocampus, embedded into a dialogue with prefrontal cortex for attention control, is known to be critical for the acquisition of declarative memories.[82,157] However, there is increasing evidence that this axis and the hippocampus also contributes to acquisition of procedural memories when encoded explicitly and when encoded implicitly with context associations.[38–40] In emotional memory formation, hippocampal activity becomes modulated via the amygdala.[45,46] All these memories, declarative, procedural and emotional, are known to preferentially benefit from subsequent sleep. Keeping temporal order in episodic memory is a function commonly attributed to hippocampal networks.[68,158] Temporal order in a sequence of events is likewise strengthened by sleep. The prefrontal cortex is particularly activated during acquisition of difficult tasks and involved also in integrating reward contingencies.[159,160] A tagging of memories via prefrontal-hippocampal circuitries during encoding could thus explain the preferential consolidation of more difficult materials and behaviors associated with expected reward during sleep. For declarative memories retrieval procedures recruiting the hippocampus in addition to prefrontal cortex, i.e., free recall and cued recall, have consistently been found to reveal greater sleep-dependent improvements than recognition procedures relying on neocortical structures. The hippocampus in conjunction with prefrontal cortex may thus establish a key circuitry that at encoding enables the subsequent sleep-dependent consolidation of a memory. Consolidation during subsequent sleep likely relies on a dialogue within the same circuitry such that the slow oscillations of SWS originating primarily from prefrontal cortex stimulate the reactivation of hippocampal memories[17] whereby these memories, in a system consolidation process, become redistributed to neocortical and other networks.[98,146]

Practice points

- Poor sleep or insufficient sleep time can substantially impair memory formation and performance levels. In the long run, learning achievements especially in children, students, the elderly and patients with sleep-related disorders can be optimized by minding their sleep habits.
- Sleep deprivation after learning hinders the consolidation of previously encoded memories. Depriving trauma victims from sleep the first night after the traumatic event might decrease the consolidation of traumatic information, potentially preventing the development of posttraumatic stress disorder (PTSD).
- Short naps are almost as beneficial for memory consolidation as a whole night of sleep. Thus, naps provide an effective and easy way to boost learning success in various areas of application.
- Learning and recapitulating learned materials right before going to bed might enhance the beneficial effect of sleep on respective memories.

Research agenda

- It remains an intriguing puzzle whether sleep strengthens non-selectively all previously acquired memory traces or if only certain designated memory traces gain access to sleep-associated consolidation. Research has to enlighten if these are especially the memories that are salient and associated with expected reward in future situations.
- Learning can differ greatly in various vital aspects, it can be intentional or incidental, implicit or explicit, activate declarative or procedural memory systems and the associated differing neuronal circuitry. Accordingly, sleep-dependent consolidation processes might substantially differ between memory tasks. Comparisons between memory tasks ought to be more explicitly considered in future research.
- Recent evidence indicates that declarative and procedural memory systems do not act as independently as originally thought but rather interact during encoding and subsequent consolidation processes. This interaction can give rise to synergies, competition and interference between these memory systems during sleep-dependent consolidation, which needs to be elaborated.
- The failure to find improved memory performance after post-learning sleep can be due to insufficient behavioral output measures, in particular when the sleep-associated system consolidation process leads to qualitative changes in the brain representation. An important issue of future research will be to establish behavioral performance measures of retrieval that more sensitively reflect sleep-dependent changes in the neuronal representation.
- Sleep stages are complex phenomena with only some of the underlying processes specifically linked to memory processing. Their specific roles for memory consolidation can be enlightened only by research concentrating on the identification of such specific neurochemical and electrophysiological events rather than on manipulating sleep stages as a whole.
- For memory consolidation, sleep is required to occur within a certain time window after learning. However the exact boundaries of this time window are presently unclear.
- Future research on sleep and memory consolidation needs to consider a broader range of subject populations. Especially investigations of different age groups and infants and children appear to be promising to unravel the psychological processes critical to sleep-dependent memory consolidation.

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